Experiment 5 – Wireshark Packet Capture and Analysis

1. Aim

To capture and analyse network packets using the **Wireshark packet capture tool**, apply filters for different protocols such as TCP, UDP, ARP, DNS, HTTP, IP, ICMP, and DHCP, and inspect the captured packets.

2. Description

Wireshark is a network analysis and packet capture tool used to capture packets in real time and display them in a human-readable format. It can be used to inspect network traffic, analyse protocol details, apply filters, troubleshoot network problems, and study network communication.

Wireshark captures packets sent and received by a computer. The captured packets can be viewed and analysed using the Packet List, Packet Details, and Packet Bytes panes.

3. Packet Capture

The network interface was selected in Wireshark, and packet capturing was configured to stop automatically after **100 packets**. The captured packets were then saved for further analysis.

The packets were filtered to display specific protocols such as:

- TCP/UDP
- ARP
- DNS
- HTTP
- IP/ICMP
- DHCP

Flow graphs were also generated for TCP/UDP and DNS packet communication.

4. Observation

The packet capture was performed using Wireshark to analyse the network communication between devices. Wireshark can operate in **promiscuous mode**, which allows the network interface to capture packets passing through the network, including packets that are not directly addressed to the computer.

ARP packets do not contain a transport layer header because ARP does not use transport layer protocols such as TCP or UDP. ARP is used to determine the MAC address associated with an IP address.

DNS mainly uses **UDP** as its transport layer protocol and commonly uses **port number 53**. TCP may also be used in certain situations.

The default port number used by the **HTTP protocol** is **80**.

A broadcast IP address is used to send packets to all devices within a particular network. For example, in the network **192.168.1.0/24**, the broadcast address is **192.168.1.255**. The student observation topics in the manual cover promiscuous mode, ARP headers, DNS transport protocol, HTTP port number, and broadcast IP addressing.

5. Outcome

Network packets were successfully captured and analysed using Wireshark. Different protocol packets were filtered and inspected, and the communication flow between devices was observed using the available packet analysis features.

6. Challenges Faced

- Selecting the correct network interface for packet capture.
- Identifying the required protocol packets among a large number of captured packets.
- Applying the correct filters for TCP, UDP, ARP, DNS, HTTP, ICMP/IP, and DHCP packets.
- Understanding the different fields present in the captured packets.
- Generating and interpreting the flow graph.
- Capturing the required number of packets successfully.

7. Result

Thus, network packets were successfully captured and analysed using the **Wireshark packet capture tool**. Different protocol packets were filtered, inspected, and analysed successfully.